

EE 605: Error Correction Codes

Problem Set 2

1. Prove that the following claims are equivalent:
 - (a) A code $\mathcal{C}$ is MDS.
 - (b) Every subset of k columns of the generator matrix G are linearly independent.
 - (c) Every subset of $n - k$ columns of H are linearly independent.
 - (d) If the generator matrix G is of the form $[I_k | A]$ ($k \times (n - k)$ matrix) where I_k is a $k \times k$ identity matrix, then A has the property that its every square submatrix has full rank.
2.
 - (a) Show that all binary (n, k, d) MDS codes are either of the form $(n, n, 1)$ or $(n, 1, n)$ or $(n, n - 1, 2)$.
 - (b) Also construct a binary MDS code of each form for any n .
3. Let $A(n, d)$ denote the largest number of codewords M in an (n, M, d) binary code.
 - (a) Show that $A(n, d) \leq 2A(n - 1, d)$.
 - (b) Show that $A(2d, d) \leq 4d$. Hint: Use the above together with the Plotkin bound derived in class for $d < n/2$.
 - (c) Show that the Reed-Muller code $RM(m, 1)$ satisfies the above with equality.
4. *Punctured codes.* Consider an (n, M, d) code $\mathcal{C}$ over a field F with minimum distance $d \geq 2$. For some fixed $i \in \{1, 2, \dots, n\}$, let $\mathcal{C}'$ be the code constructed by puncturing $\mathcal{C}$ at the i -th coordinate. Show that $\mathcal{C}'$ is an $(n - 1, M, d')$ code with

$$d' = \begin{cases} d - 1 & \text{if } \exists \text{ a minimum weight codeword } \neq \mathbf{0} \text{ in } \mathcal{C} \\ & \text{with non-zero element at } i\text{-th coordinate} \\ d & \text{otherwise} \end{cases}$$
5. (*Expurgated Hamming Codes*) Show that the dimension of the Expurgated Hamming Code is $k - 1$.
6. (*Reed-Muller codes*). Show that for any $m \geq 2$, the $(2^m, 2^m - m - 1, 4)$ extended Hamming code over $GF(2)$ is equivalent to the binary Reed-Muller code $RM(r = m - 2, m)$.
7. Consider the first-order Reed-Muller code $RM(1, 3)$.
 - (a) What is the minimum distance of the code?
 - (b) Determine all its parity-check sums and devise a majority-logic decoder for the code.

(c) Decode the received vector $\mathbf{r} = (01000101)$.

8. (*Combination of codes*) For $i = 1, 2$, let C_i be a binary (n, k_i) linear code with generator matrix G_i and minimum distance d_i , respectively. Assume that $d_2 > d_1$. We form the following linear code of length $2n$:

$$\begin{aligned} C &= [C_1 \mid C_1 + C_2] \\ &= \{[\mathbf{u} \mid \mathbf{u} + \mathbf{v}] : \mathbf{u} \in C_1 \text{ and } \mathbf{v} \in C_2\}. \end{aligned}$$

(a) C is a binary $(2n, k_1 + k_2)$ linear code. Construct a generator matrix for this code.

(b) Find the minimum distance of this code.

9. Write out a parity-check matrix of the Reed-Muller code $RM(2, 5)$.
10. Let $k \geq 1, n > d \geq 2$ and consider an $[n, k, d]$ linear code $\mathcal{C}$. Show that from $\mathcal{C}$, one can construct codes with the parameters $[n - 1, k - 1, d]$, $[n - 1, k, d - 1]$, $[n, k - 1, d]$, and $[n, k, d - 1]$.